Grade 6 Accelerated
Unit 15
Lesson 5 Practice

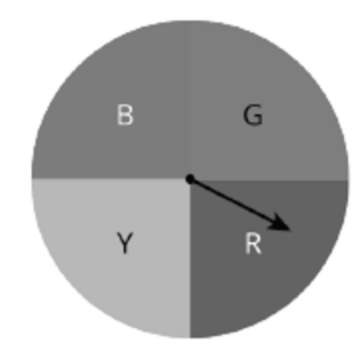
Name_Answer Key________
Date___________________
Period__________________

1. Explain the difference between theoretical and experimental probability of a simple experiment.

Theoretical is $\frac{\text{outcome}}{\text{total events in the sample space}}$.

Experimental is $\frac{\text{times the event occurs}}{\text{total number of trials}}$.

Jennifer performed a simulation and spun the arrow on the spinner below. She recorded his results in the frequency table below.



Color	Frequency
Blue	8
Green	3
Red	7
Yellow	4

2. What is the theoretical probability of landing on green?

$\frac{1}{4}$

3. What is the experimental probability of landing on green?

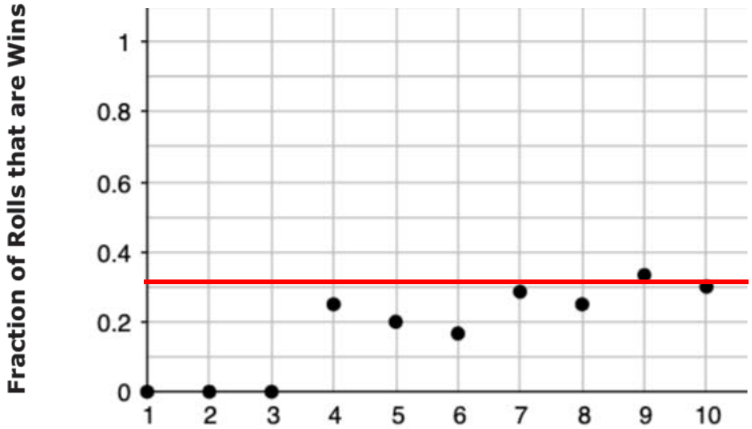
$\frac{3}{8+3+7+4} = \frac{3}{22}$

Tyler is playing a game where he wins if he rolls a 2 or a 5 on a standard number cube.

4. What is the theoretical probability that Tyler wins the game?

$\frac{2}{6}$ or $\frac{1}{3}$

He then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

   0.3

6. What type of probability is this?

   Experimental

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 2 or a 5.

8. After 10 rolls, what fraction of the total rolls were a win?

   $\frac{3}{10}$

9. If Tyler rolled the number cube 10 more times, estimate the probability that he will roll a 2 or a 5.

   $\frac{1}{3}$

10. If he were to roll the number cube 500, how many times would you expect him to roll a 2 or a 5.

    $\frac{1}{3}(500) \approx 167$ times